

Villanova University
Department of Mechanical Engineering
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FINAL REPORT 5. SOLAR THERMAL ENERGY STORAGE DESIGN

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1. Introduction and Background

Solar energy systems rely on the ability to capture and store energy, as solar input is not constant. Parabolic trough collectors were one way to efficiently concentrate incoming solar radiation into a working fluid; however, energy production was limited to periods of sunlight. To ensure continuous thermal energy, excess heat needed to be transferred into a storage medium during collection periods. Later, this stored heat energy was retrieved when solar input was unavailable. In this system, a closed-loop configuration circulated a working fluid (phase-changing Freon) through the solar collectors. The Freon absorbed thermal energy through the collector tubes and exited as a saturated vapor before delivering it to a thermal energy storage unit.

The thermal storage system considered in this report consisted of a packed bed of solid spherical pellets housed within an insulated tank. As the hot working fluid passed through the bed, heat was transferred to the pellets, which gradually increased their temperature. During this charging phase, the fluid underwent a phase change back to a saturated liquid as it released latent heat to the thermal storage unit. Once charged, the thermal storage unit's energy was extracted by passing air through vertical flow passages in the tank. This allowed the ambient air around the thermal storage unit to be heated for space heating applications. The primary design objective was to maximize both the total energy stored and the system's ability to deliver sufficiently high-temperature air over an eight-hour discharge period. This would ensure reliable thermal performance during periods without solar input.

Table 1. System Design Constraints

Constraints	Values
Time Exposed to Incident Solar Energy	6 hours
Incoming Incident Solar Heat Flux	800 W/m ²
Number of Solar Trough Collectors	5
Orientation of Solar Trough Collectors	Parallel
Length of Solar Trough Collectors	10 m
Collector Working Fluid	Freon (R-134a)
Collector Fluid Exit Quality	1.0
Thermal Storage Unit Height	5 m
Minimum Thermal Storage Unit Final Temperature	50°C
Thermal Storage Unit Thickness Ratio (B/b)	8.0
Thermal Storage Unit Structure	Solid Spherical Pellets

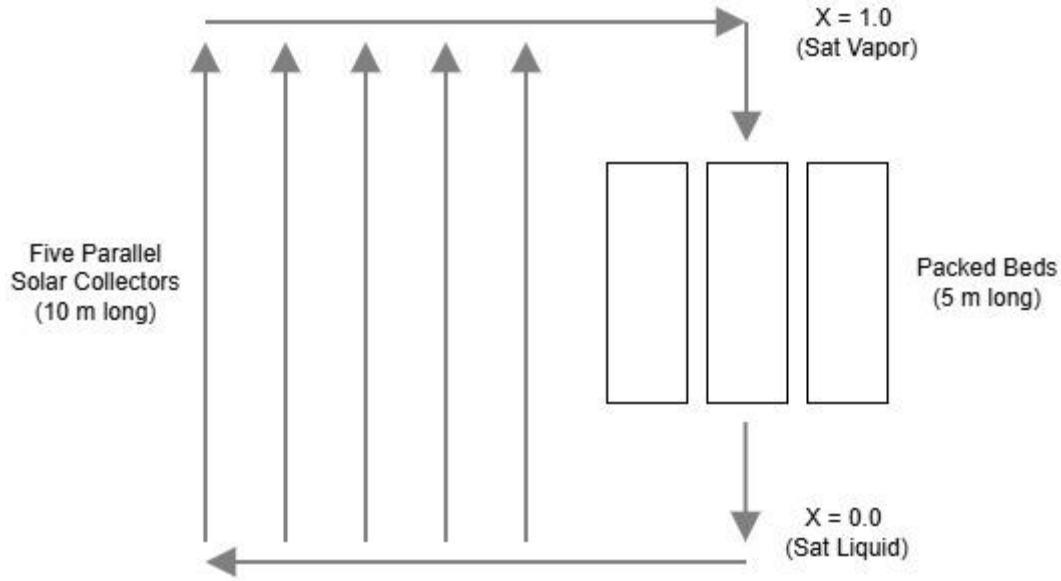


Figure 1. System Diagram

Table 2. Designed System Parameters

Parameters	Values
Solar Collector Tube Diameter	37 mm
Solar Collector Surface Properties	$\alpha = 0.94, \varepsilon = 0.04$
Freon Volumetric Flow Rate	75 L/min
Solar Collector Fluid Exit Temperature	85°C
Solar Collector Fluid Exit Pressure	2.92 MPa
Thermal Storage Unit Length Scale (B)	4.5 m
Storage Unit Pellet Material	Alumina (Al_2O_3)
Storage Unit Pellet Diameter	0.1 m

2. Packed Bed Heat Transfer

The packed bed transferred energy between the working fluid and the solid spheres to store heat in the unit. The temperature and quality of the fluid changed with position and time as the heat bed absorbed energy as it approached steady state (fully charged state).

$$U = \left(\frac{k_{hom}}{D_p}\right)(2.0 * 1.1 * Pr^{\frac{1}{3}} * Re^{0.6}) \quad (1)$$

The homogeneous k value (k_{hom}) and the porous diameter (D_p) were calculated using Equations 2 and 3. The homogeneous k calculation used the conductivity of the saturated liquid (k_f) and the saturated vapor (k_v), and the quality of the saturated mixture (X). The porous diameter calculation used the hydraulic diameter (D_H) and the porosity (ε).

$$k_{hom} = \left(\frac{X}{k_g} + \frac{1-X}{k_f} \right)^{-1} \quad (2)$$

$$D_p = \frac{3}{2} * D_H * \frac{1-\varepsilon}{\varepsilon} \quad (3)$$

Once the overall heat transfer coefficient was calculated, it was used in Equation 4 to find the temperature at each node in the packed bed. The theory behind the equation was a step model, using the current solid temperature (T_s^k) to then predict the values at the next step based on several other parameters, such as the time change (Δt), the density (ρ), the specific heat (c_p), the porosity (ε), and the surface area density (β). The time steps were given as k and $k+1$. The fluid temperature was T_f , and the solid temperature was T_s .

$$T_s^{k+1} = T_s^k + \left(\frac{U * \beta * \Delta t}{\rho * c_p * (1-\varepsilon)} \right) * (T_f - T_s^k) \quad (4)$$

The next calculation found the quality of the fluid at each node in the packed bed. Some of the parameters from Equation 4 were similar, but Equation 5 also used the change in position between nodes (Δx), the cross-sectional area (A_c), the mass flow rate of the fluid ($\dot{m}$), and the enthalpy of the fluid (h_{fg}). The quality was itemized using position steps of $i-1$ and $i+1$ (before and after the current node).

$$X_{i+1}^{k+1} = X_{i-1}^k - (2 * \Delta x) * \left(\frac{U * \beta * A_c}{\dot{m} * h_{fg}} \right) * (T_f - T_s^k) \quad (5)$$

The EES code experimented with different pellet diameters to determine an optimal size. The material was selected as Al_2O_3 (Alumina) after evaluating the optimal specific heat of the solid, which was around 0.9 kJ/kg. Alumina fit this constraint ($c_p \sim 0.88$ to 0.92) and was considered an industry-standard material.¹

3. Trough Collector Calculations

Solar energy was collected by a parabolic trough collector that concentrated the radiation to a collector tube. Five collectors were operating in parallel, each 10 meters long. For this report, the design evaluated the tube diameter, the flow rate, and the fluid temperature (the saturation temperature of the refrigerant). The system was designed by first focusing on one tube, then scaling the design choices to all five parallel pipes.

One design point that had to be addressed was material. Research showed that a common material for solar collectors was stainless steel, to offer high corrosion resistance and no chemical reaction with the refrigerant, unlike some materials such as copper.² Additionally, it was necessary to add a coating to the outer surface of the collector. The chosen coating needed high absorptivity to capture as much radiation as possible, while also having a low emissivity to prevent heat from escaping. Research showed that titanium nitride oxide (TiNOX) was an industry standard material, with an absorptivity of ~ 0.94 and an emissivity of ~ 0.04 .³

The next step in the collector design was to determine the solar heat flux that the system was designed to operate in (Denver, Colorado, during the summer hours). The average daily solar power in June was researched to be 6.3 kWh/m².⁴ Over a six-hour solar collection period, this power converted to 1.05kW/m². There was also a 50 to 1 concentration ratio given in the initial constraints, so the total heat flux on the collector tube was 52.5kW/m². The actual heat rate into the pipe was found by Equation 6, using the total heat flux (q''), the diameter of the tube (d_t), and the length of the tube (L_t). The heat flux only affected the bottom half of the tube.

$$q = q'' * \frac{\pi * d_t}{2} * L_t \quad (6)$$

The next step was to determine how much energy was needed to completely change the refrigerant from a saturated liquid to a saturated vapor. This was dependent on the mass flow rate ($\dot{m}$) and the enthalpy of vaporization (h_{fg}).

$$q = \dot{m} * h_{fg} \quad (7)$$

$$d_t = \frac{2 * \dot{m} * h_{fg}}{q'' * L_t} \quad (8)$$

The mass flow rate was a design input, and the enthalpy of vaporization was dependent on the fluid temperature (also a design point). Based on these three factors, an operating point was selected. The mass flow rate was 15 L/min, and the saturation temperature was selected as 85°C (at 29.4 bar). The saturation temperature was selected to be safely below the critical temperature of 101°C. The resulting diameter of the collector tube was 37 mm to ensure full phase change of the fluid through the collectors.

The outer surface temperature was calculated by modeling the pipe wall as a resistance network. The fluid was on one end, and the outer surface was the other. The convection with the wall and the conduction through the wall were the two resistors in this model. The calculation for the outer surface temperature (T_w), shown in Equation 9, used q , the heat through the wall, and R , the total resistance.

$$T_w = T_{sat} + q * R \quad (9)$$

The resistance (R) was calculated as a convection resistor and a conduction resistor. The total resistance calculation was shown in Equation 10, where t was the wall's thickness, U_t was the convective heat transfer coefficient, k was the wall's conduction coefficient, and A was the total surface area of the pipe. The tube was chosen to be steel with a conductivity of around 15 W/m-K, and the wall thickness was 3 mm.

$$R = \frac{t}{k * A} + \frac{1}{U_t * A} \quad (10)$$

The convective heat transfer coefficient calculation was complex because the fluid was experiencing boiling as it went through the collector tube. The Gungor and Winterton approximation was used. Equation 11 was a function of the boiling number (Bo), the quality (X),

the densities of the fluid as a liquid and as a vapor (ρ_l, ρ_v), and the liquid convective heat transfer coefficient (U_l).

$$U_t = U_l * (1 + 3000 * Bo^{0.86} + 1.12 * \left(\frac{X}{1-X}\right)^{0.75} * \left(\frac{\rho_l}{\rho_v}\right)^{0.41}) \quad (11)$$

With the total resistance, the outer surface temperature was calculated to be 96.3°C. This made sense, since the Freon temperature was not very high compared to other fluids that could be used in solar applications. Also, the outer temperature should have remained below the critical temperature of Freon so that the fluid could not overheat and lose its saturated quality.

4. EES Code Plots

The packed bed was modeled in EES as a looped system to solve for the temperature and quality of the solid bed at various heights and timesteps. The bed was divided into six nodes, each one solved based on the temperature and quality of the previous node, as modeled by the equations from Section 2 of this report. Figure 2 shows the P-h diagram for the fluid moving through the packed bed as it transitions isobarically from a saturated vapor to a saturated liquid at the packed bed outlet.

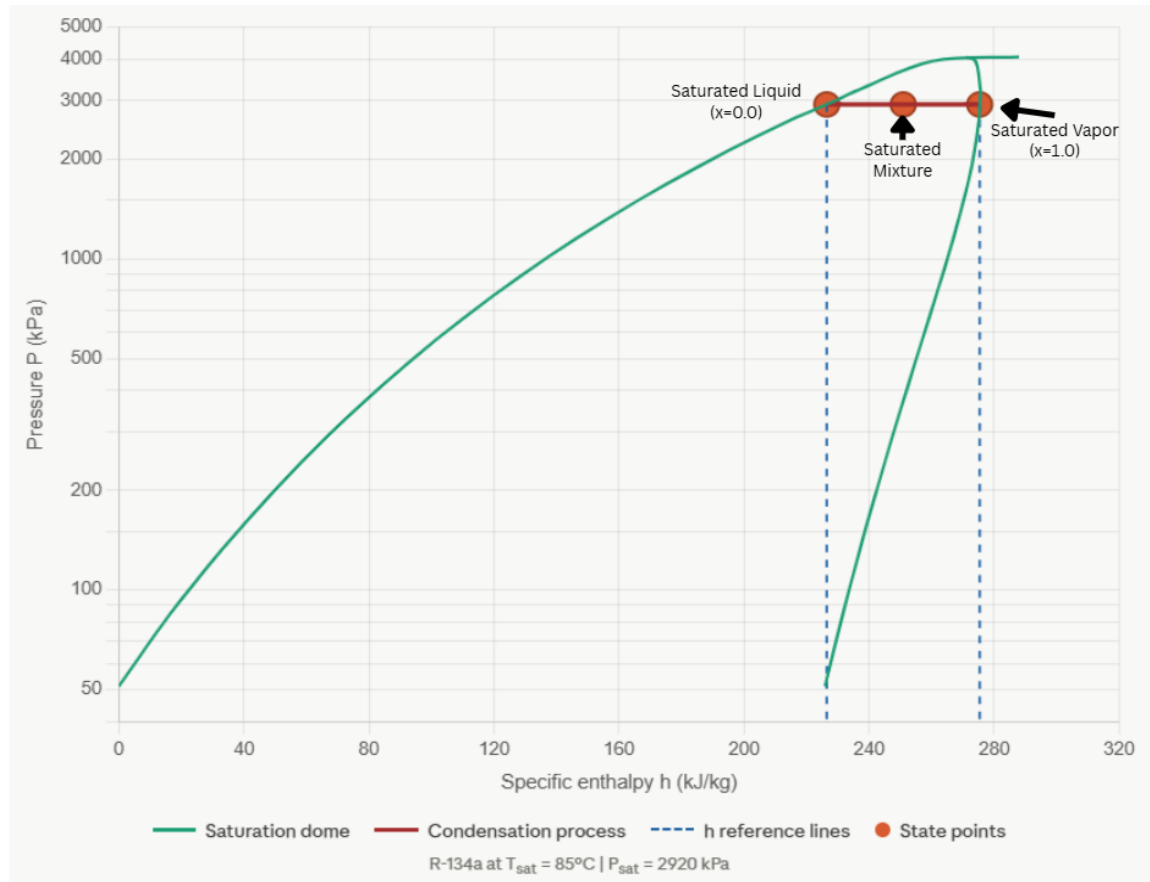


Figure 2. System P-h Diagram

The main output of the EES code was the thermal storage unit temperature at each node and time. Figure 3 shows the temperature curves as a function of height and 30-minute intervals (increasing left to right). The EES software solved the system equations and directly output a table of the data. This EES-generated data table was then fed into Claude AI and prompted to plot the tabulated data in graphical format, resulting in Figure 3.

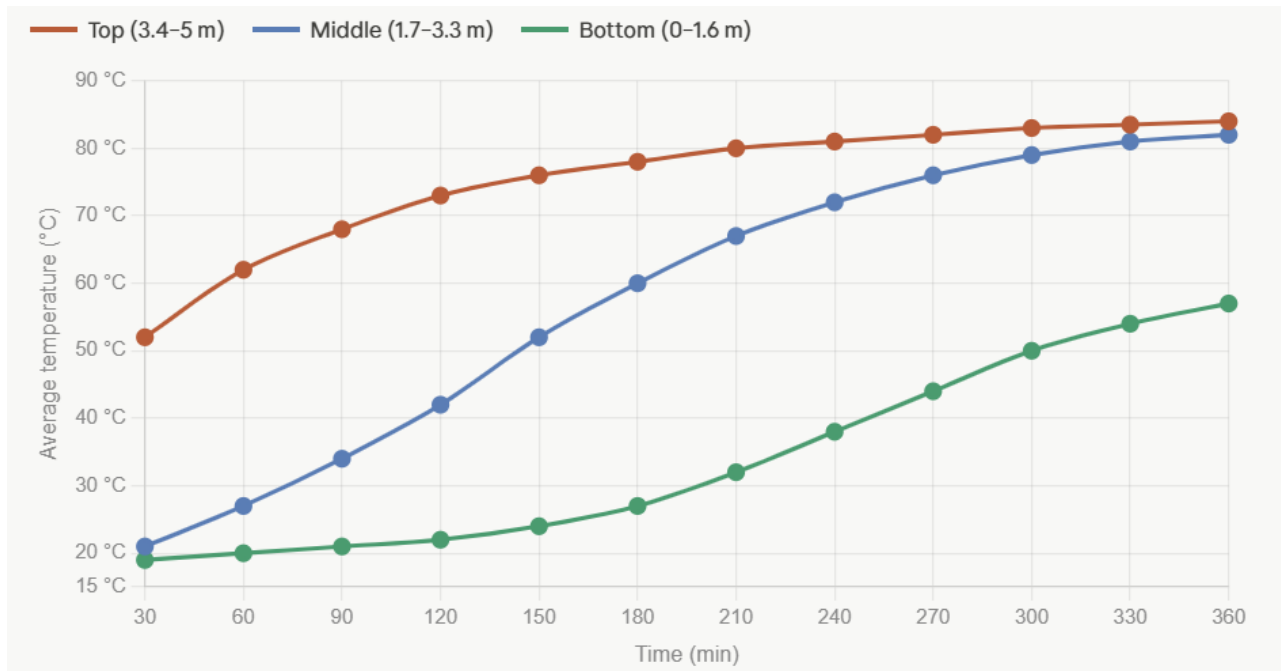


Figure 3. Average Thermal Storage Unit Temperature by Sections during Solar Collection Period

Figure 4 was created by feeding the same EES data table into Claude AI and prompting it to make a transient positional graph similar to the example sketch provided in the class lecture notes. The plots generated consisted of best-fit lines through temperature points at vertical intervals of 1 meter. The lines differed by 30-minute intervals.

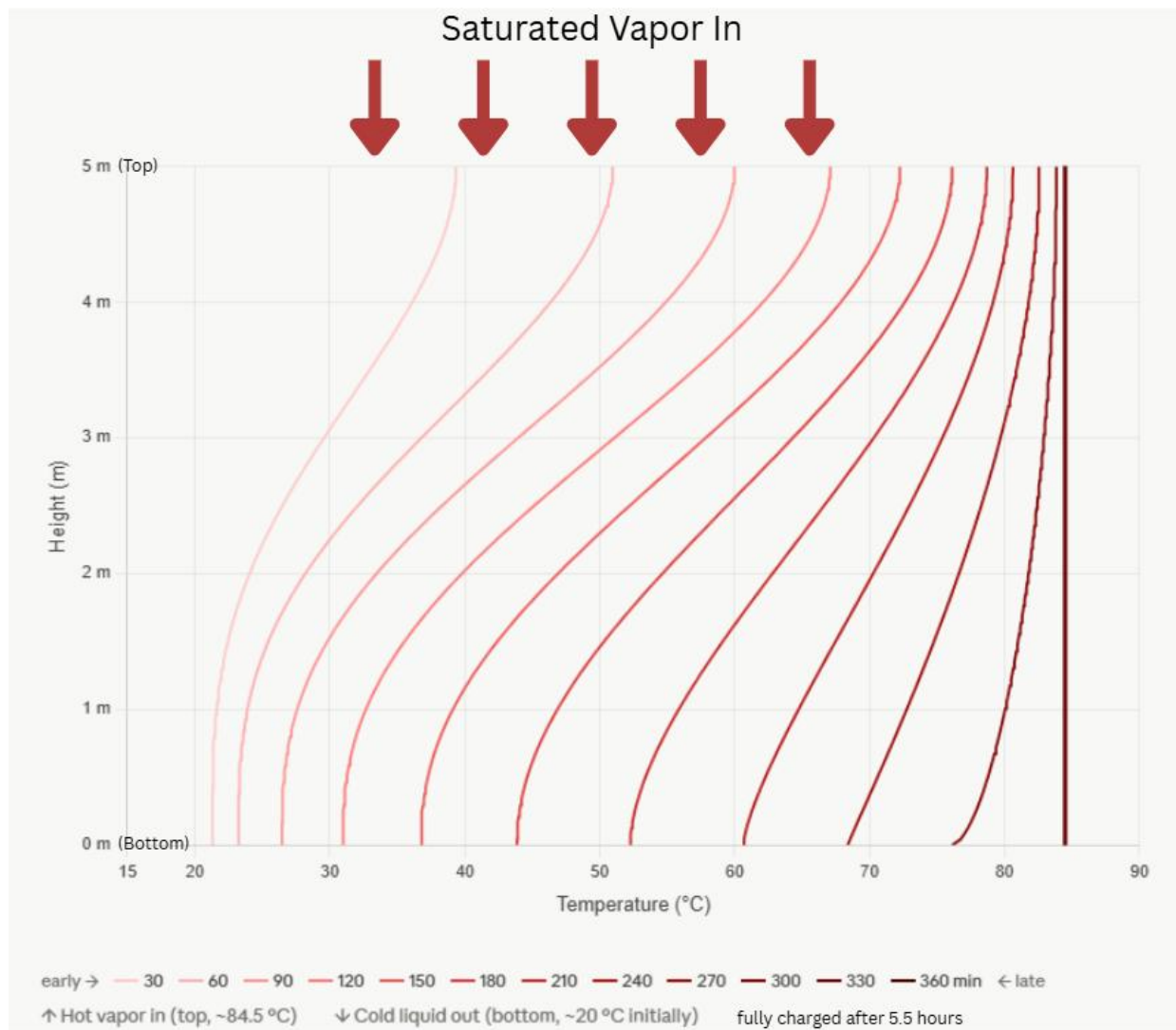


Figure 4. Thermal Storage Unit Transient Positional Temperature Diagram

Once the transient positional temperature plot for the thermal storage unit was generated, the data was used to create a more intuitive visual representation. This inspired the generation of the transient heat map for the thermal storage unit, which yielded a numerically equivalent visual shown differently in Figure 5.

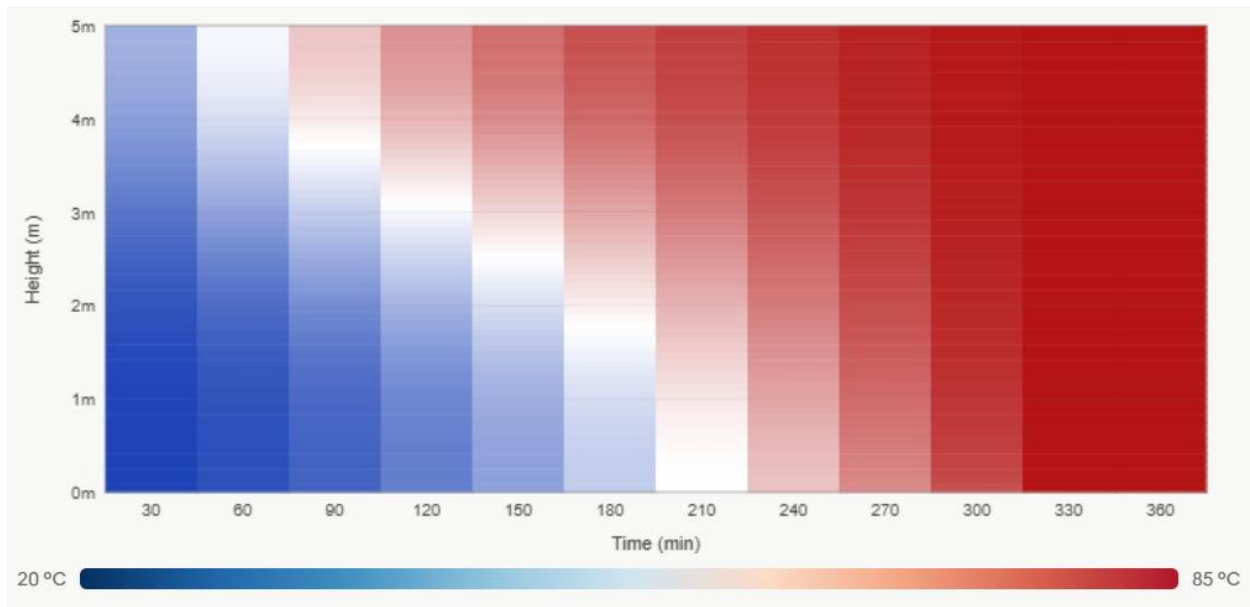


Figure 5. Thermal Storage Unit Heat Map over Time

To visualize how energy was stored in the thermal storage unit during its 6-hour solar collection period, the EES system code included thermal energy variables. One of these variables represented thermal energy stored in the pack bed unit. This variable was included in the EES duplicate function loops to generate a table of transient data. The tabulated data, when fed to Claude AI, resulted in Figure 6.

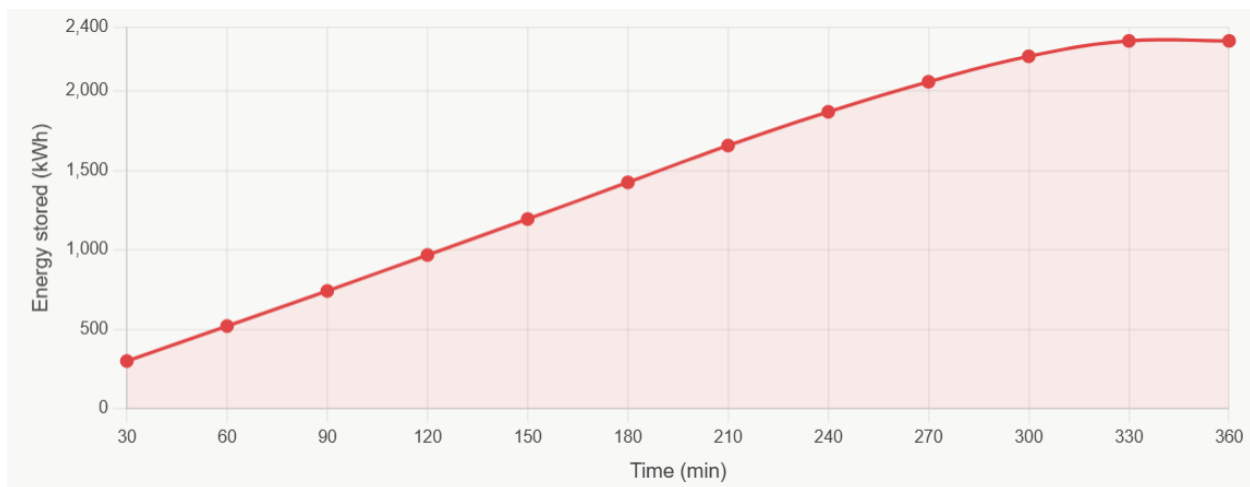


Figure 6. Total Thermal Energy Stored over Time

Figures 7 and 8 were created to examine the Freon quality throughout its system circulation. The plots were generated using the EES quality tables as a function of position. These plots were then laid on the system component background photos to visualize the Freon's phase change throughout the components.

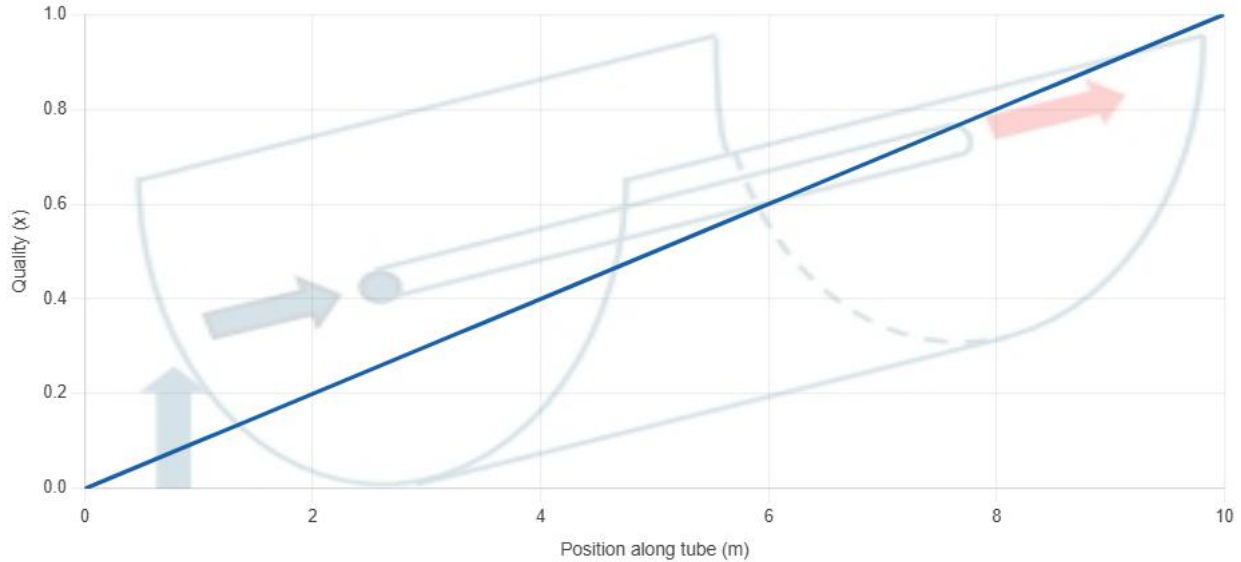


Figure 7. Freon Quality versus Position along Collector Tubes

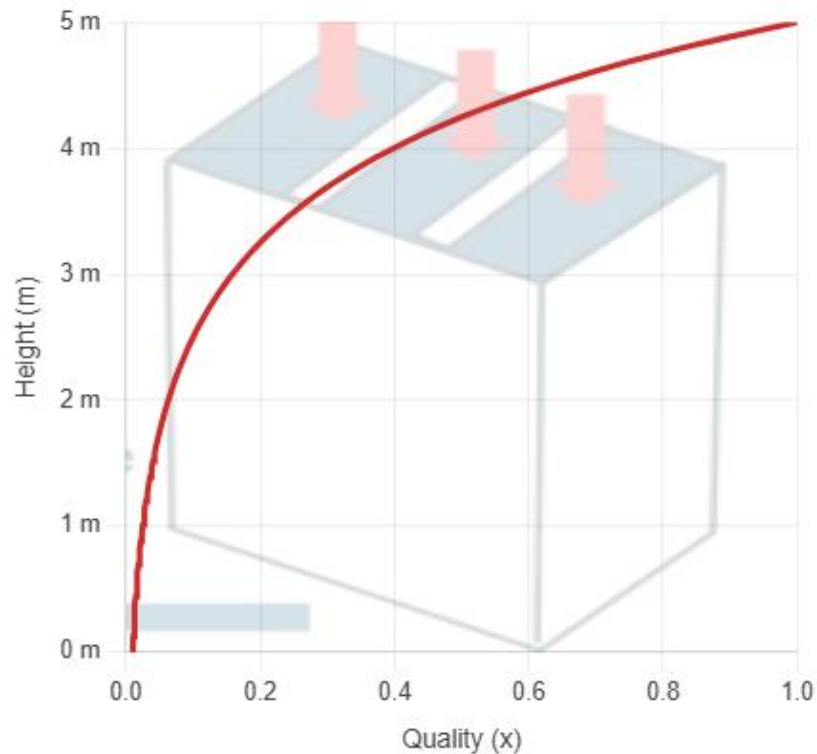


Figure 8. Thermal Storage Unit Height versus Freon Quality

5. Conclusion

The final packed bed design reaches equilibrium temperature within six hours of direct energy input during the day in Denver, Colorado. The heat flux was amplified by a factor of 50 by the trough reflectors into the stainless-steel pipe coated with a titanium nitrate oxide material. The refrigerant was heated from a saturated liquid to a saturated vapor in the collector, then cooled back to a saturated liquid in the packed bed, transferring and storing the energy in the solid medium.

The rate of energy input into the packed bed was linear until about the fourth hour, when the temperatures became nearly equal, and the rate of change slowed. This made sense based on the mechanics of convection, as the rate of energy transfer was an effect of the temperature difference between the fluid and the solid.

Another interesting outcome of this project was the quality change through the packed bed. Figure 8 showed that the quality assumed a radical curve, with rapid change at the beginning of the bed, and slower change towards the bottom. This could have been because the hot vapor was shedding all of its energy early on, with the energy rate slowing as the vapor cooled off. This also showed why the temperature at the bottom of the bed took longer to reach equilibrium.

The system temperature (85°C) seemed low for these applications, considering discussions held in class, but this seemed reasonable considering the working fluid. Freon (R-134a) had a critical temperature of 101°C, and a very low boiling point of around -30°C. With that in mind, 85°C was very high for the fluid used and seemed reasonable considering the high specific heat, allowing it to transfer as much energy as needed from the solar collector to the packed bed.

Overall, this project demanded learning and research into two-phase heat transfer, boiling convection, packed beds, and solar systems in general. From evaluating the effects of vapor bubble formation through the collector tube to the interaction of the refrigerant with the spherical solid matrix in the packed bed, this was the most complex heat transfer equation of the semester, which made this report challenging yet extremely interesting.

References

- (1) *Our Technology*. Alumina Energy. <https://www.aluminaenergy.com/technology> (accessed 2026-05-02).
- (2) Gretz, W. Parabolic-Trough Solar Water Heating.
- (3) *TiNOX*. Almeco Group. <https://almecogroup.com/en/surface/tinox/> (accessed 2026-05-02).
- (4) *Solar Energy and Solar Power in Denver, CO*. Solar Energy Local. <https://www.solarenergylocal.com/states/colorado/denver/> (accessed 2026-04-29).

Appendix

A. Engineering Equation Solver

```
"=== GEOMETRY & DISCRETIZATION ==="
N      = 10                {spatial nodes in packed bed}
M      = 12                {time steps}
t_total = 21600 [s]        {6-hour charging window}
dt      = t_total / M      {= 1800 s per step}

"=== SOLAR COLLECTOR ==="
N_coll = 5                 {number of parallel tubes}
L_coll = 10 [m]            {tube length}
D_tube = 0.037 [m]         {tube inner diameter}
W_aper = 1.5 [m]           {aperture width per collector}

A_tube = pi * D_tube^2 / 4 {tube cross-sectional area [m^2]}
P_perim = pi * D_tube      {FIX: full circumference, not half [m]}

"--- Solar flux on tube wall ---"
"q_flux = absorbed power per unit of heated perimeter length [kW/m^2]"
"G_sol [kW/m^2] * eta_opt * W_aper [m] => linear power density [kW/m]"
"Divide by P_perim [m] to get wall heat flux [kW/m^2]"
G_sol = 0.900 [kW/m^2]
eta_opt = 0.75
q_flux = G_sol * eta_opt * W_aper / P_perim {absorbed flux [kW/m^2]}
Q_tube = q_flux * P_perim * L_coll           {power per tube [kW]}
Q_total = Q_tube * N_coll                    {total collector power [kW]}

"=== PACKED BED GEOMETRY ==="
L_bed = 5.0 [m]
B_dim = 4.5 [m]
A_bed = B_dim^2 {bed cross-sectional area [m^2]}
dx     = L_bed / N {spatial step [m]}

"=== PELLET PROPERTIES (concrete/stone) ==="
D_part = 0.10 [m] {particle diameter - chosen so tau >> dt}
eps     = 0.36 {void fraction}
beta_sa = 6 * (1 - eps) / D_part {volumetric surface area [m^2/m^3]}
rho_s   = 2200 [kg/m^3]
Cp_s    = 0.900 [kJ/kg-K]
```

"=== R-134a SATURATION PROPERTIES AT T_sat ==="

T_sat = 85 [C]

P_sat = P_sat(R134a, T=T_sat)

h_f = enthalpy(R134a, T=T_sat, x=0)

rho_f = density(R134a, T=T_sat, x=0)

mu_f = viscosity(R134a, T=T_sat, x=0)

k_f = conductivity(R134a, T=T_sat, x=0) {W/m-K}

Cp_f = Cp(R134a, T=T_sat, x=0) {kJ/kg-K}

Pr_f = Prandtl(R134a, T=T_sat, x=0)

h_g = enthalpy(R134a, T=T_sat, x=1)

rho_g = density(R134a, T=T_sat, x=1)

mu_g = viscosity(R134a, T=T_sat, x=1)

k_g = conductivity(R134a, T=T_sat, x=1) {W/m-K}

Cp_g = Cp(R134a, T=T_sat, x=1) {kJ/kg-K}

h_fg = h_g - h_f {latent heat [kJ/kg]}

"=== MASS FLOW RATE FROM ENERGY BALANCE ==="

"Each tube absorbs Q_tube [kW]; fluid condenses so Q = m_dot * h_fg"

m_dot_1 = Q_tube / h_fg {per-tube mass flow [kg/s]}

m_dot = m_dot_1 * N_coll {total mass flow [kg/s]}

V_dot_Lmin = m_dot / rho_f * 1000 * 60 {volumetric flow [L/min]}

"=== COLLECTOR TUBE: DITTUS-BOELTER (liquid baseline) ==="

"mu_f from EES is in [kg/m-s]; k_f in [W/m-K] -> convert to [kW/m-K]"

Re_f = 4 * m_dot_1 / (pi * D_tube * mu_f)

Nu_f = 0.023 * Re_f^0.8 * Pr_f^0.4

h_l = Nu_f * (k_f / 1000) / D_tube {liquid HTC [kW/m^2-K]}

"=== GUNGOR-WINTERTON TWO-PHASE HTC AT x=0.5 ==="

x_mid = 0.5

G_mass = m_dot_1 / A_tube {mass flux [kg/m^2-s]}

Bo = q_flux / (G_mass * h_fg) {Boiling number [-]}

h_tp_GW = (1 + 3000 * Bo^0.86 + 1.12 * (x_mid / (1 - x_mid))^0.75 * (rho_f / rho_g)^0.41) * h_l

T_wall = T_sat + q_flux / h_tp_GW {tube wall temperature [C]}

"=== PACKED BED: HOMOGENEOUS TWO-PHASE PROPERTIES AT $x_{avg}=0.5$ ==="

"Harmonic-mean conductivity and viscosity; volume-fraction mean density"

$x_{avg} = 0.5$

$k_{hom} = 1 / (x_{avg} / (k_g / 1000) + (1 - x_{avg}) / (k_f / 1000))$ {kW/m-K}

$\rho_{hom} = 1 / (x_{avg} / \rho_g + (1 - x_{avg}) / \rho_f)$ {kg/m³}

$\mu_{hom} = 1 / (x_{avg} / \mu_g + (1 - x_{avg}) / \mu_f)$ {kg/m-s}

"FIX: mixture Cp weighted by mass quality (not pure liquid Cp)"

$Cp_{hom} = x_{avg} * Cp_g + (1 - x_{avg}) * Cp_f$ {kJ/kg-K}

"Hydraulic pore diameter for packed bed Re"

$D_{pore} = (2/3) * D_{part} * (eps / (1 - eps))$

"Superficial velocity based on total flow and void fraction"

$u_{bed} = m_{dot} / (\rho_{hom} * eps * A_{bed})$ {interstitial velocity [m/s]}

$Re_{bed} = \rho_{hom} * u_{bed} * D_{pore} / \mu_{hom}$ {pore Reynolds number}

"FIX: Prandtl uses homogeneous Cp, not pure liquid"

$Pr_{hom} = \mu_{hom} * Cp_{hom} / k_{hom}$ {homogeneous Prandtl number}

"Wakao-Kaguei Nusselt correlation for packed beds"

$h_{int_bed} = (k_{hom} / D_{part}) * (2.0 + 1.1 * Pr_{hom}^{1/3} * Re_{bed}^{0.6})$ {kW/m²-K}

"=== FD MODEL PREFACTORS ==="

$T_f = T_{sat}$ {fluid temperature = saturation temperature throughout two-phase region [C]}

"C_s: rate constant for solid node approaching T_f [1/s]"

$C_s = h_{int_bed} * beta_{sa} / (\rho_s * Cp_s * (1 - eps))$

"C_X: spatial quality gradient coefficient [1/m] per unit (T_f - Ts) [K]"

"Derived from: $m_{dot} * h_{fg} * dX/dx = -h_{int} * beta_{sa} * A_{bed} * (T_f - Ts)$ "

"Sign: fluid flows from i=1 (inlet, x=0) to i=N (exit, x=L_bed)"

"Condensing fluid LOSES quality from inlet to exit => $dX/dx < 0$ when $Ts < T_f$ "

$C_X = h_{int_bed} * beta_{sa} * A_{bed} / (m_{dot} * h_{fg})$ {[1/(m*K)]}

```

Duplicate k = 0, 12
X[1, k] = 1.0
X[2, k] = X[1, k] - 2 * dx * C_X * (T_f - Ts[1, k])
X[3, k] = X[2, k] - 2 * dx * C_X * (T_f - Ts[2, k])
X[4, k] = X[3, k] - 2 * dx * C_X * (T_f - Ts[3, k])
X[5, k] = X[4, k] - 2 * dx * C_X * (T_f - Ts[4, k])
X[6, k] = X[5, k] - 2 * dx * C_X * (T_f - Ts[5, k])
X[7, k] = X[6, k] - 2 * dx * C_X * (T_f - Ts[6, k])
X[8, k] = X[7, k] - 2 * dx * C_X * (T_f - Ts[7, k])
X[9, k] = X[8, k] - 2 * dx * C_X * (T_f - Ts[8, k])
X[10, k] = X[9, k] - 2 * dx * C_X * (T_f - Ts[9, k])
End

"=== OUTPUT TEMPERATURES FOR PLOTTING (every 30 min) ==="
Duplicate k = 0, 12
T_inlet[k] = Ts[1, k] {bed inlet node temperature [C]}
T_middle[k] = Ts[5, k] {bed midpoint temperature [C]}
T_exit[k] = Ts[10, k] {bed exit node temperature [C]}
t_hr[k] = k * dt / 3600
End

"=== FINAL RESULTS AT END OF 6-HOUR CHARGING PERIOD ==="
T_min_final = Ts[1, 12] {coldest node (inlet) at t=6hr [C]}
T_max_final = Ts[10, 12] {hottest node (exit) at t=6hr [C]}
T_avg_final = (Ts[1,12]+Ts[2,12]+Ts[3,12]+Ts[4,12]+Ts[5,12] +Ts[6,12]+Ts[7,12]+Ts[8,12]+Ts[9,12]+Ts[10,12]) / N

M_bed = rho_s * (1 - eps) * A_bed * L_bed {mass of solid [kg]}
E_stored = M_bed * Cp_s * (T_avg_final - T_s_init) {energy stored [kJ]}
E_stored_kWh = E_stored / 3600 {energy stored [kWh]}

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Figure A-1. EES Code